Business Problem Statement

A leading retail company aims to gain deeper insights into customer shopping behavior to enhance sales performance, improve customer satisfaction, and build long-term loyalty. Recent observations indicate shifts in purchasing patterns across different demographics, product categories, and sales channels (online vs. offline).

The analysis focuses on identifying the key factors influencing customer decisions, such as discounts, product reviews, seasonal trends, and payment preferences, as well as understanding drivers of repeat purchases.

This project addresses the following core business question:

"How can consumer shopping data be leveraged to identify trends, enhance customer engagement, and optimize marketing and product strategies?"

Project Objectives & Deliverables

This project is structured to simulate a real-world end-to-end data analytics workflow, covering the following components:

1. Data Preparation & Modeling (Python)

- Clean and preprocess raw data

- Handle missing values and inconsistencies

- Perform data transformation and feature engineering

2. Data Analysis (SQL)

- Structure data into a relational format

- Simulate business transactions

- Execute SQL queries to analyze customer segments, loyalty, and purchase behavior

3. Visualization & Insights (Power BI)

- Develop an interactive dashboard

- Highlight key trends, patterns, and KPIs

- Enable data-driven decision-making for stakeholders

4. Report & Presentation

- Summarize key findings and insights

- Provide actionable business recommendations

- Present results in a clear and visually compelling format